Hick Fitts Miller Laws Analysis

This study was produced with AI support for interface prototypes and calculation templates.

SECTION 1: Persona and Scientific Foundation

Persona: Elif Nurse (32), 7-year Emergency Department Triage Nurse

Elif Nurse must determine a triage code (Red/Yellow/Green) for each patient in an average of 45-60 seconds in an environment with high cognitive load, time pressure, and noise. Fast and accurate decision-making is critical in the emergency department environment.

Academic Reference:
"Cognitive Load and Decision Making in Emergency Department Triage: An Empirical HCI Study" (Alireza et al., Emergency Medicine & Human Factors Journal).

Note: No real patient data or existing hospital interface images are used in the system; all data is synthetic/mockup data.

SECTION 2: Hick Law Analysis

Hick Law: Decision time is proportional to the logarithm of the number of choices.
Formula: $T = b * \log_2(n + 1)$

Bad Design (hick_bad.html)

There are 20 triage complaint buttons arranged irregularly on the screen that are difficult to distinguish. The user must examine each button separately.

The user spends approximately 791 milliseconds to make a selection. This time is too long in a critical emergency!

Good Design (hick_good.html)

Progressive Disclosure method is used. First, 4 main systems are selected, then relevant sub-options are shown. User directly selects the correct category, reducing decision time significantly.

Number of choices reduced from 20 to 4. Decision time reduced from 791 ms to 418 ms! Cognitive load significantly decreased with ~47% time improvement.

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SECTION 3: Fitts Law Analysis

Fitts Law: Movement time depends on target distance and target width.
Formula: MT = a + b * log2(2D/W)

Bad Design (fitts_bad.html)

The critical Confirm Triage Code and Emergency Call button is located in the top right corner of the screen, 16px x 16px in size and 900px away from the user.

Movement time is approximately 967 milliseconds. The button is too small and far away! Critical delay risk in emergency!

Good Design (fitts_good.html)

The critical button is located in the bottom control band of the screen, 280px x 54px in size and only 120px away from the user.

Movement time reduced from 967 ms to 124 ms! ~87% improvement! Button enlarged and positioned close to user.

SECTION 4: Miller Law Analysis

Miller Law: Human working memory has a capacity of approximately 7 plus or minus 2 units. Ungrouped information pieces increase cognitive load.

Bad Design (miller_bad.html)

15 mixed parameters are stacked in a single form. No grouping or hierarchy exists.

The user must process 15 different information pieces at the same time! This situation significantly increases cognitive load and causes data entry errors.

Good Design (miller_good.html)

15 mixed parameters are grouped into 3 main logical cards. Each card contains at most 4-5 information pieces.

Information pieces are separated into logical groups (chunks)! Each card is within Miller Law limit! The user only has to

deal with 4 information pieces at the same time.

SECTION 5: Trade-off Analysis

Research Question: Can Hick, Fitts and Miller laws achieve their best state together on the same screen, or does one get satisfied while the other breaks?

Trade-off Analysis:

1. Fitts vs. Miller: When we make buttons huge to satisfy Fitts Law, the area they occupy on the screen increases. This situation narrows the area of other data on the screen and disrupts Miller Law chunking balance.

2. Fitts vs. Hick: When we optimize Hick Law by dividing steps into parts (progressive menu), the total number of clicks and total movement time required for Fitts increase.

3. Hick vs. Miller: Hiding choices (progressive disclosure) as required by Hick Law can complicate Miller Law chunking structure and make information access difficult.

Defended Trade-off Decision:

In the emergency department context, Fitts Law is prioritized for vital actions. Emergency confirmation buttons are always accessible and large. Miller Law is preserved to zero out cognitive errors in the data verification stage - chunked data cards are used. For general navigation, Hick Law is sacrificed to progressive disclosure for space savings.

Conclusion: The three laws cannot achieve their best state at the same time. In one context (emergency), Fitts is prioritized, while in another context (data entry), Miller is preserved. The designer must establish the appropriate balance according to the context requirements.

CONCLUSION

This study analyzed the application of Hick, Fitts and Miller laws in an emergency department triage system. Through bad and good design examples, the effect of each law on user experience was shown mathematically. The trade-off analysis revealed that the three laws cannot achieve their best state at the same time, but an appropriate balance can be established according to the context.

For emergency department workers like Elif Nurse, Fitts Law should be prioritized for vital actions, Miller Law should be

preserved in data verification stages, and Hick Law should be sacrificed to progressive disclosure for general navigation.